

Hodge type Igusa stacks and cohomology of Shimura varieties

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Igusa stacks are geometric objects akin to Shimura varieties. Scholze predicted their existence, motivated by questions related to local-global compatibility in the Langlands program. In [2], the author has constructed Igusa stacks for some PEL-type Shimura data. The current project [1] extends this to Hodge type Shimura data. This geometric result enables us to apply the geometric local Langlands correspondence of [3] to study the ℓ -adic cohomology of Shimura varieties.

More precisely, let (G, X) be a Hodge type Shimura datum with Hodge cocharacter μ and reflex field E . Fix a prime number p , a place v of E above p and set $E = E_v$, $G = G_{\mathbb{Q}_p}$. We consider the Shimura variety attached to (G, X) over E , i.e. a tower of algebraic varieties $\{\mathbf{Sh}_K\}_K$ for K running through neat compact open subgroups of $G(\mathbb{A}_f)$. Fixing K^p , we denote by $\mathbf{Sh}_{K^p}$ the inverse limit $\lim_{K^p \rightarrow 1} \mathbf{Sh}_{K^p K_p}$. Let $\ell \neq p$ be a prime that does not divide the order of $\pi_0(Z_G)$, where Z_G is the center of G . We consider a torsion noetherian $\mathbb{Z}_\ell$ -algebra Λ containing a fixed square root of p . Our main cohomological results are as follows:

Theorem 1 (Mantovan's product formula). *There exists a filtration on $R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda)$ by complexes of smooth representations of $G(\mathbb{Q}_p) \times W_E$, labeled by the Kottwitz set $B(G, \mu^{-1})$, whose graded pieces are given by*

$$R\Gamma(\mathrm{Ig}^b, \Lambda)^{\mathrm{op}} \otimes_{\mathcal{C}_c(G_b(\mathbb{Q}_p), \Lambda)}^{\mathbb{L}} R\Gamma_c(G, b, \mu).$$

Here Ig^b denotes a perfect Igusa variety over $\overline{\mathbb{F}}_p$; G_b is an inner form of a Levi of G determined by b , and up to shifts and twists, $R\Gamma_c(G, b, \mu)$ is the cohomology of a local Shimura variety attached to the datum (G, b, μ) . This describes the structure of $R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda)$ and extends results of [4, 6]. Compared to [5], we do not make extra assumptions on the Hodge type Shimura data.

Let $W_E \subset \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ be the Weil group of E and fix the prime-to- p level K^p .

Theorem 2 (Eichler-Shimura relation). *Assume G is unramified, and fix a Iwahoric in a hyperspecial subgroup $I_p \subset K_p \subset G(\mathbb{Q}_p)$. Let $\mathcal{H}_{K_p}$ be the corresponding spherical Hecke algebra with Λ coefficient. Then the inertia subgroup of W_E acts unipotently on $R\Gamma(\mathbf{Sh}_{I_p K^p, \overline{E}}, \Lambda)$, and for any lift σ of the Frobenius, the relation $H_\mu(\sigma) = 0$ holds, where $H_\mu(X) \in \mathcal{H}_{K_p}[X]^1$ is a renormalized Hecke polynomial.*

This result generalizes the classical Eichler-Shimura relation for modular curves, confirming Blasius-Rogawski's conjecture [9, Section 6] for Hodge type Shimura varieties and extends the work of [10]. It shows that at all unramified primes, the Frobenius eigenvalues are constraint to the roots of the Hecke polynomials, and thus provide much information about the cohomology of the Shimura variety (over E) as a global Galois representation. Note that most previous works only consider

¹where $\mathcal{H}_{K_p}$ identifies with the center of $\mathcal{H}_{I_p}$ via Bernstein isomorphism

Shimura varieties at hyperspecial level. It is therefore a novelty that we establish this result at Iwahoric levels. Our proof is strongly influenced by [7, 8].

Theorem 3 (“Generic” part). *Assume G is unramified and “well-understood”², K_p is hyperspecial and $\mathbf{Sh}_K$ is compact. Then for a generic L -parameter ϕ , the isotypical part $R\Gamma(\mathbf{Sh}_{K,\overline{E}}, \Lambda)_\phi$ ³ is concentrated in the middle degree.*

Here we use the notion of “generic” in the sense of [11], which is inspired by [12, 13, 14]. This result shows analogy to the phenomenon that tempered automorphic representations only occur in the middle degree in the cohomology of locally symmetric spaces attached to compact Shimura varieties, albeit with torsion coefficients. Our approach follows and extends [6].

All the above results are consequences of our main geometric result below. To our surprise, given this geometric input, the cohomological consequences follow formally from the setup of [3], with a minimal amount of extra effort. One therefore wonders how much more the categorical local Langlands correspondence can be exploited to provide information about the global Langlands correspondence, as a first approximation of which, about the cohomology of Shimura varieties.

We write Perf_v for the v -site of perfectoid spaces over $\overline{\mathbb{F}}_p$. The (good reduction locus of the) Shimura variety will be considered as a sheaf on Perf_v and denoted by $\mathbf{Sh}_{K^p}^{\diamond, \circ}$. Let Bun_G be the stack of G -bundles on the Fargues-Fontaine curve with its Beauville-Laszlo map $BL : \mathrm{Gr}_G \rightarrow \mathrm{Bun}_G$, from the B_{dR}^+ -affine Grassmannian for G . We restrict BL to the minuscule Schubert cell $\mathrm{Gr}_{G, \mu^{-1}}$.

Theorem 4. *There exists a functorial (in Shimura data) construction of an “Igusa stack” Igs_{K^p} on Perf_v , together with a Cartesian diagram*

$$\begin{array}{ccc} \mathbf{Sh}_{K^p}^{\diamond, \circ} & \xrightarrow{\pi_{\mathrm{HT}}^{\diamond}} & \mathrm{Gr}_{G, \mu^{-1}} \\ \downarrow & & \downarrow BL \\ \mathrm{Igs}_{K^p} & \xrightarrow{\overline{\pi}_{\mathrm{HT}}} & \mathrm{Bun}_{G, \mu^{-1}}, \end{array}$$

where the top row is the Hodge-Tate period map. Furthermore, for all $\ell \neq p$, Igs_{K^p} is ℓ -cohomologically smooth of dimension 0 with constant dualizing sheaf.

Let us give some ideas how this is used to prove Theorem 1 to 3. Theorem 1 is direct: $\mathrm{Bun}_{G, \mu^{-1}}$ has a Newton stratification labeled by $B(G, \mu^{-1})$, and the theorem follows by identifying the stratum corresponding to $b \in B(G, \mu^{-1})$ of the Cartedian diagram. In particular, horizontally the fiber is the Igusa variety Ig^b while vertically the fiber is the local Shimura variety attached to (G, b, μ) . For Theorem 2 and 3, we consider the complex

$$\mathcal{F} := R\overline{\pi}_{\mathrm{HT},*} \Lambda \in D(\mathrm{Bun}_{G, \mu^{-1}}, \Lambda).$$

It is not hard to check that there is a $G(\mathbb{Q}_p) \times W_E$ -equivariant isomorphism

$$i_1^* T_\mu \mathcal{F}[-d](-\frac{d}{2}) \simeq R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda),$$

²As defined in David Hansen’s talk, see the corresponding abstract in this proceeding.

³Defined as in [6, Section 4].

where T_μ is the Hecke operator attached to μ , and $i_1 : \underline{BG}(\mathbb{Q}_p) \hookrightarrow \text{Bun}_G$ is the open immersion of the neutral stratum. This reduces the problem of studying $R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda)$ to studying $\mathcal{F}$ and the functor $i_1^* T_\mu$ individually. Indeed, Theorem 3 follows by showing $\mathcal{F}$ is perverse for a t-structure on $D(\text{Bun}_{G, \mu^{-1}}, \Lambda)$ while $i_1^* T_\mu$ (whose target identifies with the derived category of smooth $G(\mathbb{Q}_p)$ -representations with usual t-structure) is perverse t-exact on generic objects.

Let $X_{\widehat{G}}$ be the stack of L -parameters for G . Theorem 2 uses the spectral action of [3, Section X], a Λ -linear action of the ∞ -category of perfect complexes on $X_{\widehat{G}}$ on $\mathcal{D}(\text{Bun}_G, \Lambda)$. In particular, the Hecke operator T_μ acts as (the spectral action of) a vector bundle $\mathcal{V}_\mu$ on $X_{\widehat{G}}$. This leads to an action of the endomorphism algebra of $\mathcal{V}_\mu$ on $R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda) = i_1^* T_\mu \mathcal{F}[-d](-\frac{d}{2})$. Since W_E acts on $\mathcal{V}_\mu$ via the universal L -parameter, through this it also acts on $R\Gamma(\mathbf{Sh}_{K^p, \overline{E}}, \Lambda)$. One can show that this spectral W_E -action coincides with the natural W_E -action coming from the structure map of $\mathbf{Sh}_{K^p}$ to $\text{Spec} E$. Now any Frobenius lift, as an endomorphism of $\mathcal{V}_\mu$, has a characteristic polynomial with coefficients in the spectral Bernstein center. Passing to Iwahoric levels, the latter acts on $R\Gamma(\mathbf{Sh}_{K^p I_p}, \Lambda)$ through the Iwahoric-Hecke algebra, and this polynomial reduces to the Hecke polynomial. The theorem follows.

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